

Ahlam Tarik

📍 Fez, Morocco ✉ ahlam.tarik@eidia.ueuromed.org 📞 +212-688-855563

🌐 linkedin.com/in/theahlamtarik 🐙 github.com/theahlamtarik

SUMMARY

AI Engineering student ranked **1st in class** (GPA 18.21/20, 2025–26), working on **self-supervised representation learning** for physiological and audio signals. Research internships at **KAUST** (audio watermark detection) and **École Centrale Casablanca** (atrial-fibrillation onset forecasting), with a focus on evaluation design: source-disjoint splits, shortcut controls, falsification tests.

EDUCATION

- **Euro-Mediterranean University of Fez (UEMF) – EIDIA** Fez, Morocco
Engineering Degree in Artificial Intelligence (5-year program, equiv. M.Eng.) Sep 2022 – Jun 2027
 - **UEMF**: 1st in Morocco, 2nd in Africa (RUR 2025) · **EIDIA**: first AI engineering school in Africa · **Merit Scholarship**
 - **Ranked 1st in class** · GPA **18.21/20** (2025–26)

RESEARCH & ACADEMIC EXPERIENCE

- **Research Intern – École Centrale Casablanca** Casablanca, Morocco
Supervised by Dr. Kawtar Zerhouni & Mr. Hermann Agossou – self-supervised ECG for AF onset forecasting Jun – Sep 2026
 - **Precursor or patient fingerprint?** Probing horizon-conditioned **HEPA/JEPA** representations for 5-min **AF onset forecasting**: do they encode time-local precursors, or stable patient-specific signatures?
 - Designed a patient-held-out, leakage-resistant protocol (AF-history and RR/HRV baselines, same-patient temporal controls, false-alarm analysis); **Python/WFDB** pilot on 10,699 decision points from **MIT-BIH AFDB**: AF detection **0.978 AUROC** vs HRV-baseline forecasting at chance (0.49)
- **Research Intern – KAUST, Resilient Computing & Cybersecurity Center** Remote
Supervised by Dr. Mehdi Zakroum & Prof. Ali Shoker – neural audio watermark forensics under distribution shift Jun 2026 – Present
 - Formulated a detector-side forensic task through a critical review of 30+ papers: separating watermark-removed from never-watermarked speech, targeting generalization to **unseen schemes, removal attacks and corpora**
 - Built the pipeline (**PyTorch**, **AudioSeal**, frozen **WavLM-base+**): speaker-disjoint splits and attack-matched controls ruling out content and processing shortcuts; layer-wise probing over 13 layers localizes watermark evidence in **early encoder layers** (80.9% 3-class, 91.4% attack ID)
- **Random Survival Forests for Alzheimer’s Disease Progression** UEMF – Supervised Research
Supervised by Prof. Abderazzak Mouiha, PhD, P.Stat. – ADNI clinical dataset Jan 2026 – Present
 - Modelling **MCI-to-Alzheimer’s conversion** on longitudinal ADNI data with **Random Survival Forests**, benchmarked against Cox proportional-hazards and joint longitudinal-survival models
- **T-MECA: Trustworthy Multi-Encoding Classification of Arrhythmias** UEMF
Science Award Winner (1st Prize) – Spring School (UM6P, École Polytechnique, OCP) Jan 2026
 - Four signal-to-image encodings (CWT, GAF, MTF, RP) fused with **MC-Dropout uncertainty decomposition**: **0.916 macro-F1** on the class-imbalanced MIT-BIH benchmark, rising to **0.947** when the 10% least-confident beats are referred
- **Morocco AI Summer School – Multimodal Foundation Models** INPT & IMT Nord Europe
Selected among 50 from 1,300+ applicants (top 3.8%) – Sponsored by Google & ACM Sep 2025
 - Architected **EVA**, a multimodal presentation-coaching framework (speech, LLM semantic scoring, prosody, gesture and sign-language recognition); one of 3 laureates selected to **demo at Morocco’s Women AI Forum 2026**

TECHNICAL SKILLS

Programming: Python, SQL, R, C/C++

ML/DL: PyTorch, TensorFlow/Keras, scikit-learn, scikit-survival, Hugging Face Transformers, WavLM, AudioSeal, OpenCV/YOLO

Methods: self-supervised & contrastive learning, uncertainty quantification, survival analysis, multimodal fusion, RAG

LLMs & Agents: LangChain, LangGraph, CrewAI, FAISS, Whisper **Tools**: Git, Linux, Pandas, NumPy

Languages: Arabic (native) · French (C1) · English (C1) · Spanish (B1)

INDUSTRY EXPERIENCE

- **AI Engineer Intern – Agence de Développement du Digital (ADD)** Rabat, Morocco
Morocco’s national digital development agency (Ministry of Digital Transition) Jul 2025 – Sep 2025
 - Delivered an **agentic AI assistant** (LangGraph, multi-step tool-calling, RAG) for the national “Academia Raqmya” platform
- **AI Developer – Mystery Gifts (Startup)** Remote
Multi-agent gift recommendation engine (CrewAI), exhibited at GITEX Africa 2025 & MEDays 2025 Feb 2025 – Present

SELECTED PROJECTS

- **AlWarid – AI Healthcare Decision Support Platform** RamadanIA 2026: 1st regionally, 2nd nationally
 - Voice-to-report generation, **explainable diagnostic support** under uncertainty and a multilingual patient portal (Ministry of Digital Transition); **WassIA** module – Alzheimer’s risk prediction and monitoring for elderly patients and clinicians: **Jury Favourite Prize**, Maison de l’Intelligence Artificielle hackathon; selected for the **7th UNIMED Start Cup** (Italy, Nov 2026)
- **AtlasFields – precision agriculture: Sentinel-1 SAR + agentic LLM pipeline** Act in Space Morocco (CNES) 2026 – 2nd

LEADERSHIP & COMMUNITY

Head of AI Department, Digiclub (UEMF’s first IT club) · **Local Organizer**, Hugging Face LeRobot Hackathon (sole organizer in

Morocco) · **Mentor**, Digigirlz Morocco (Microsoft) · **AI Instructor**, Algorithmics International School (K-12)

2024 – Present